

Normalization & Standardization

Standardization

Z-Score method / Formula.

Centers around the values near mean & standard deviation 1.

Z-Score Normaly.

$$\frac{x - \mu}{\sigma}$$

Normalization is affected by outliers as they have extreme value or low are compressed to sub-range.

Standardization handles outliers more nicely as it considers mean & std.

Normalization.

Putting values in common scale,

Units - Take out.

Put - common scale -

Keep - magnitude on common scale.

— min-max Scaling.

$$X_i^o = \frac{X_i - X_{min}}{X_{max} - X_{min}}$$

— we do it to get distributed scale.

→ [0, 1].

Requires
Scaling Not
Decision Tree
XG-Boost.

Scaling		Standard
KNN Linear Regress Logistic image 0-255 pixels	SVM Linear Logistic Regress	